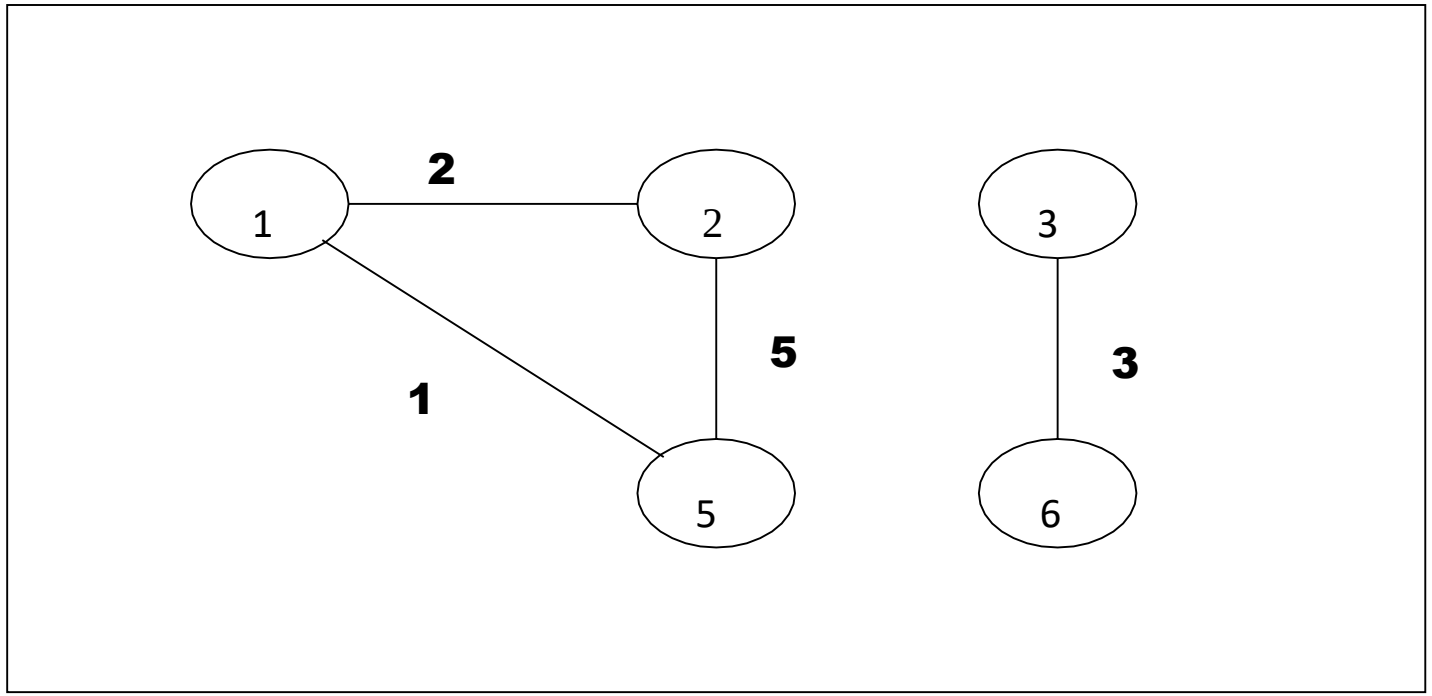
Kruskal's Algorithm

Session No.:
Course Name: Analysis and Design of Algorithms
Course Code: R1UC407B
Instructor Name: Dr. Mili Dhar

Review of the key concepts of session no.

1. Prim's algorithm is used to find the **Minimum Spanning Tree (MST)** of a connected, undirected, weighted graph.
2. The MST connects all vertices with the minimum possible total edge weight, without forming cycles.
3. It grows the MST from a single starting vertex by **continuously adding the smallest edge** that connects a vertex in the MST to a vertex outside the MST.

What do you think how do you find MST for a disconnected graph?



At the end of this session students will be able to

Learning Outcome:

- Apply Kruskal's Algorithm in a weighted graph.

Session Outline

1. Introduction of Kruskal’s Algorithm

2. Apply Kruskal's algorithm on a given graph

3. A hands-on approach to apply Kruskal’s method on a weighted graph

Introduction of Kruskal’s Algorithm

Kruskal's Algorithm is a **greedy algorithm** used to find the **Minimum Spanning Tree (MST)** of a connected/disconnected, undirected, weighted graph. A Minimum Spanning Tree is a subset of the graph's edges that connects all vertices with the minimum possible total edge weight, without forming any cycles.

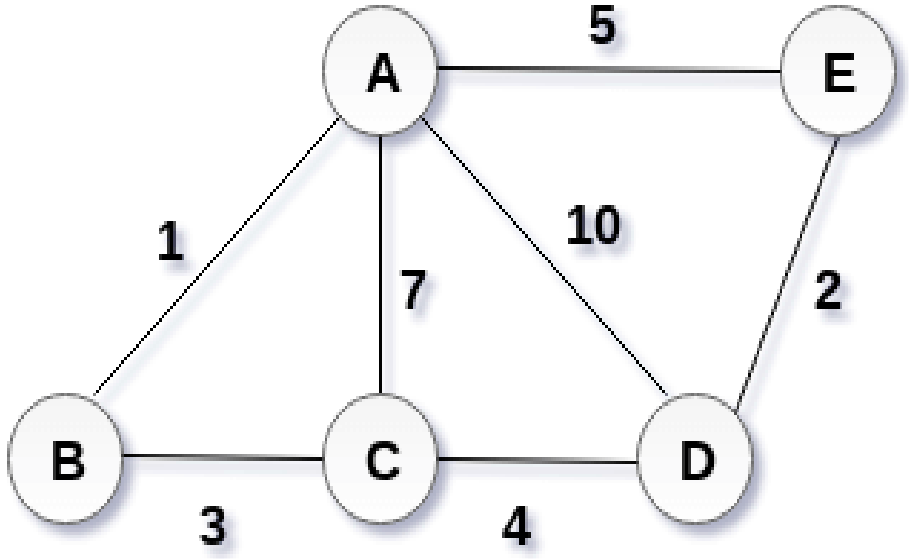
The Kruskal's algorithm is given as follows.

Algorithm

1. Sort all the edges in non-decreasing order of their weight.
2. Pick the smallest edge. Check if it forms a cycle with the spanning tree formed so far. If the cycle is not formed, include this edge. Else, discard it.
3. Repeat step#2 until there are (V-1) edges in the spanning tree.

Learning Activity- 1

Example : Apply the Kruskal's algorithm on the graph given as follows.



Solution: The weight of the edges is given as:

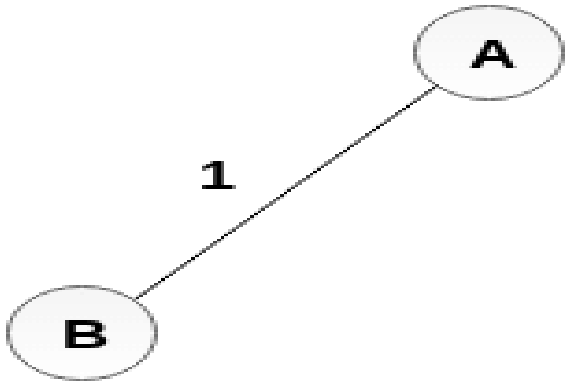
Edge	AE	AD	AC	AB	BC	CD	DE
Weight	5	10	7	1	3	4	2

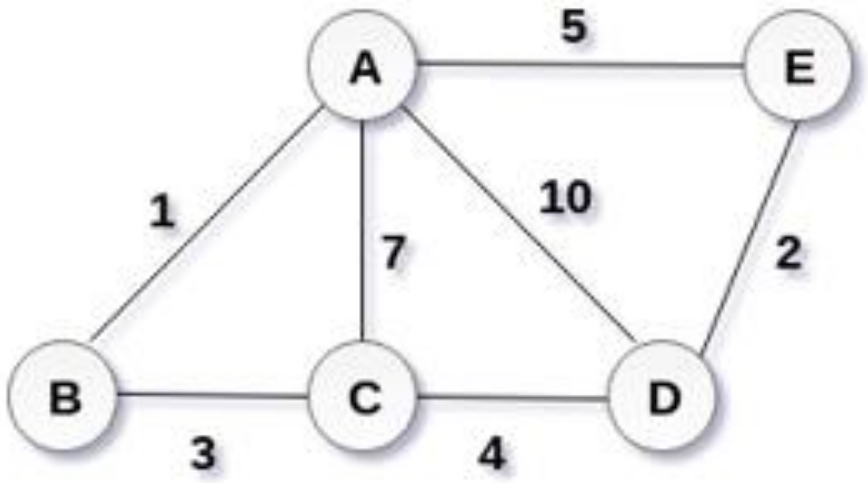
Learning Activity-1

Sort the edges according to their weights.

Edge	AB	DE	BC	CD	AE	AC	AD
Weight	1	2	3	4	5	7	10

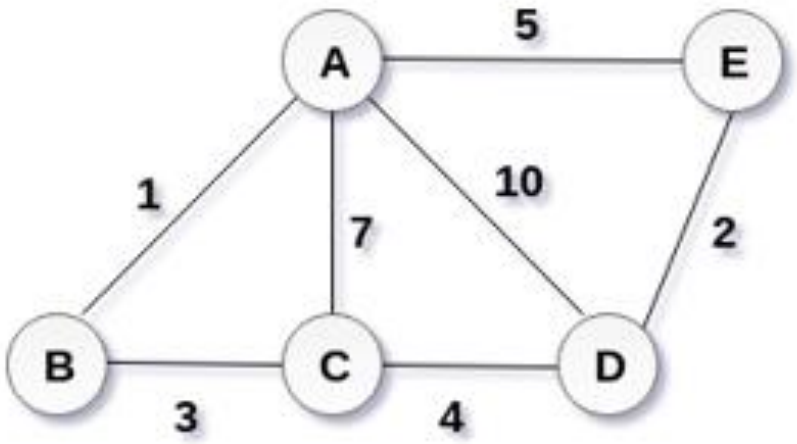
Start constructing the tree; Add **AB** to the MST;



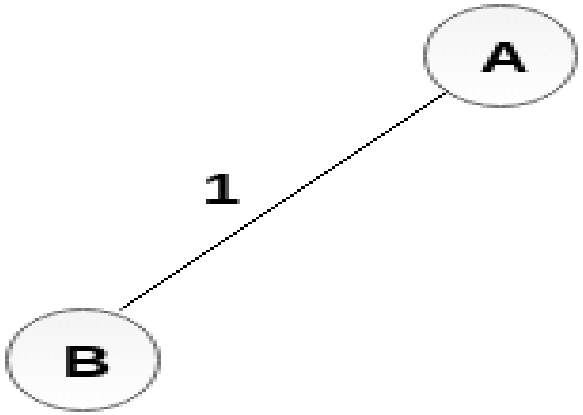


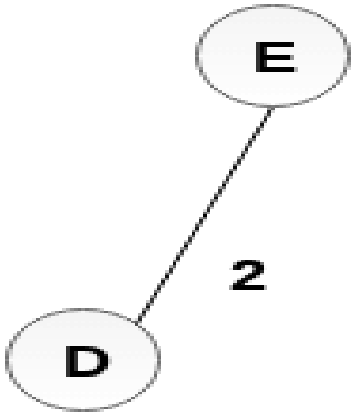
Learning Activity-1

Edge	AB	DE	BC	CD	AE	AC	AD
Weight	1	2	3	4	5	7	10



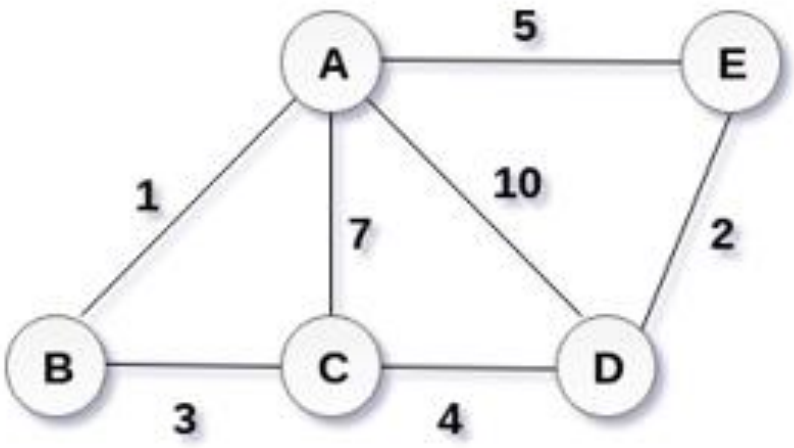
Add **DE** to the MST;



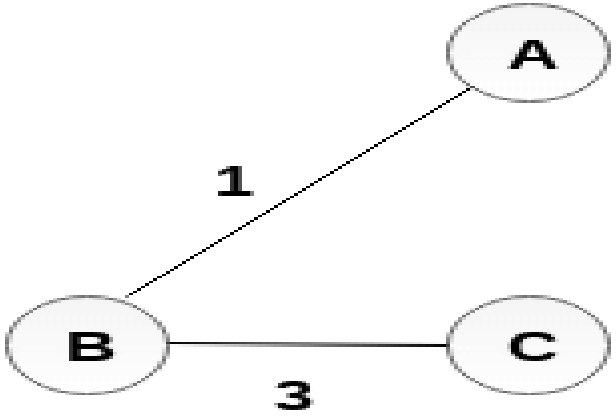


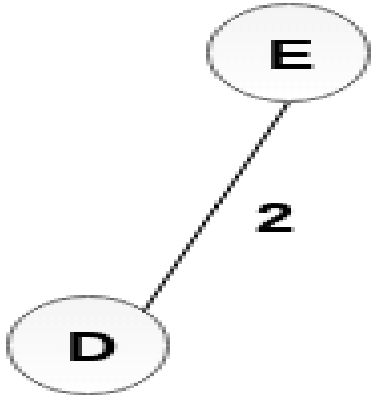
Learning Activity-1

Edge	AB	DE	BC	CD	AE	AC	AD
Weight	1	2	3	4	5	7	10



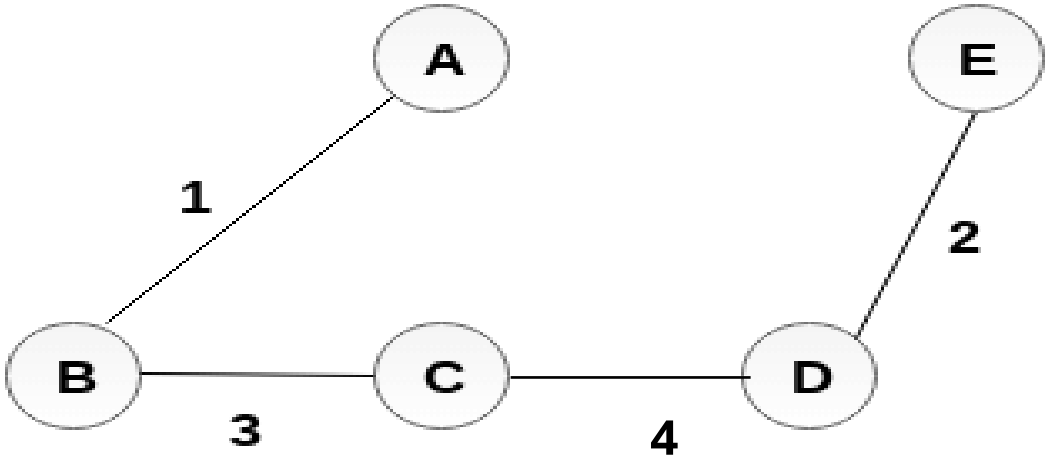
Add **BC** to the MST;





Learning Activity-1

The next step is to add **CD**.



The next step is to add **AE,** but we can't add that as it will cause a **cycle**.

The next edge to be added is **AC**, but it can't be added as it will cause a **cycle**.

The next edge to be added is **AD**, but it can't be added as it will contain a **cycle**.

Hence, the final MST is the one which is shown above.

Cost of MST = 1 + 2 + 3 + 4 = 10

Learning Activity-2

1. Go to wooclap.com
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Event code
QCBSFE

Summary

1. Finds the **Minimum Spanning Tree (MST)** of a connected, undirected, weighted graph.
2. Sorts all edges by weight and adds the smallest edge that doesn't form a cycle.
3. If there are multiple edges with the same weight, any of them can be chosen without affecting the overall MST cost.
4. Different choices for these equally weighted edges can result in different MSTs.
5. It is particularly useful for sparse graphs where the number of edges is much less than the maximum possible number of edges.

Next Session....

We will learn about the Time and Space Complexity of Kruskal's Algorithm